

Week 4 lab worksheet.

Name _____ Date _____

PhysioEx Exercise 3: Neurophysiology of Nerve Impulses, Activities 1 to 9. About 2 to 3 hours with the simulation.

Do each activity in PhysioEx first, then fill in its three boxes here while it is fresh. Write in your own words, in full sentences, and use your actual numbers.

What goes in each box

- 1. What did you measure?** Name what you changed and what you recorded, with units. For example, the stimulus voltage you set and the membrane potential you read.
- 2. What were your findings?** Give the actual results, with numbers. What went up, what went down, and what stayed the same.
- 3. What did you learn?** Explain why the results came out that way, using the physiology from this week. This is the most important box.

Your worksheet is graded once PhysioEx shows each activity complete.

Activity 1

The Resting Membrane Potential

What did you measure?

What were your findings?

What did you learn?

Activity 2

Receptor Potential

What did you measure?

What were your findings?

What did you learn?

Activity 3

The Action Potential: Threshold

What did you measure?

What were your findings?

What did you learn?

Activity 4

The Action Potential: Importance of Voltage-Gated Na⁺ Channels

What did you measure?

What were your findings?

What did you learn?

Activity 5

The Action Potential: Measuring Its Absolute and Relative Refractory Periods

What did you measure?

What were your findings?

What did you learn?

Activity 6

The Action Potential: Coding for Stimulus Intensity

What did you measure?

What were your findings?

What did you learn?

Activity 7

The Action Potential: Conduction Velocity

What did you measure?

What were your findings?

What did you learn?

Activity 8

Chemical Synaptic Transmission and Neurotransmitter Release

What did you measure?

What were your findings?

What did you learn?

Activity 9

The Action Potential: Putting It All Together

What did you measure?

What were your findings?

What did you learn?

Looking back at the whole lab

Which result surprised you most, and what did you expect instead?

Pick one activity and connect it to one competency from this week. Which one, and how does the lab show it?

Turn it in: photograph or scan these pages and upload them to the Week 4 lab assignment in Canvas.